

- 1). The company wants to automate its payroll process. write a C program to calculate the total salary by adding bonus amount

Pseudo code

START

Read basic-Salary

Read bonus

$\text{total-Salary} = \text{basic-Salary} + \text{bonus}$

Print total salary

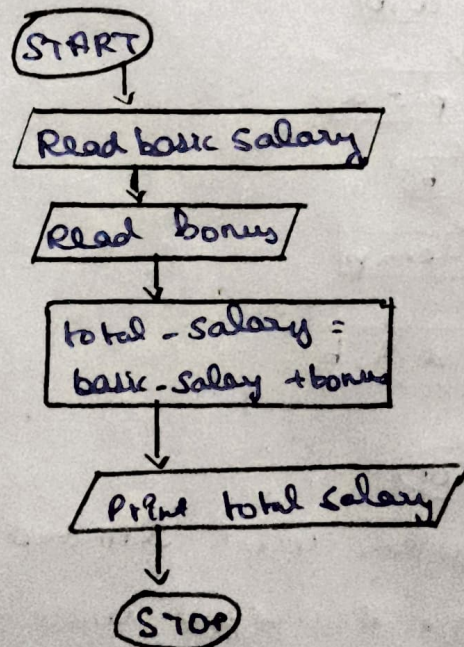
STOP

I/P:- basic-Salary = 30000

bonus = 5000

O/P:- Total.Salary = 35000

Flow chart :-



2. During a database update, two emp ID used needs to be exchanged. Write a C program to swap

Pseudo code

START

Read a, b

temp = a

a = b

b = temp

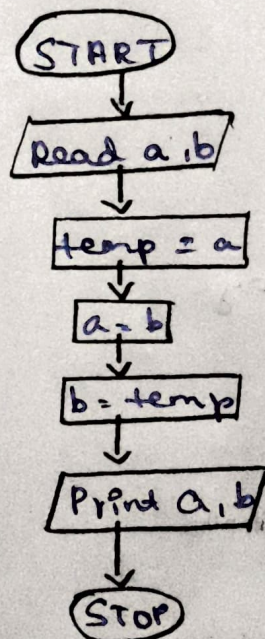
Print a, b

STOP

Input : a = 5, b = 10

Output : a = 10, b = 5

Flow chart



3. In an embedded system with limited memory, two values must be exchanged w/o using any extra storage. Write a C program

Pseudo code

START

Read a, b

$a = a + b$

$b = a - b$

$a = a - b$

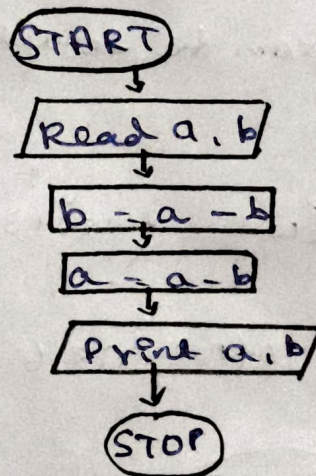
PRINT a, b

STOP

Input : $a = 5, b = 10$

Output :- $a = 10, b = 5$

Flowchart :-



- 4). A Bank wants to provide customer with an estimate of the interest & write a C program to calculate the simple interest.

START

READ P, R, T

$SI = (P \times R \times T) / 100$

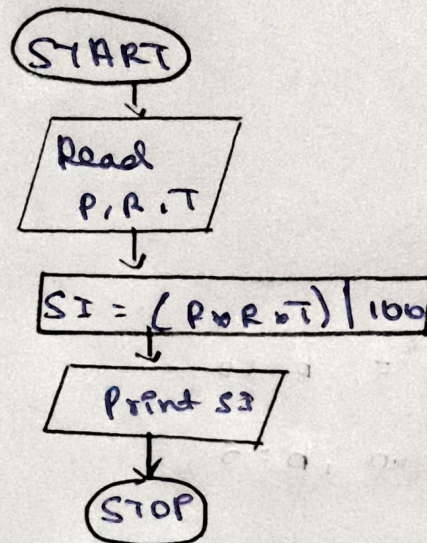
PRINT SI

STOP

Input :- $P = 10000$, $R = 5$, $T = 2$

Output :- Simple Interest = 1000

Flow chart



5). write a c program to convert C to F :

START

Read celcius

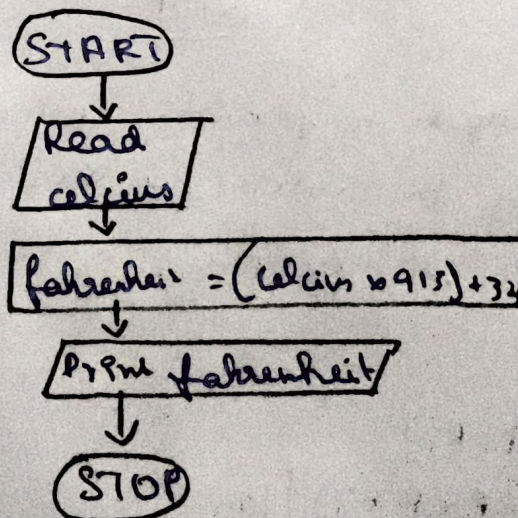
fahrenheit = (celcius * 9/5) + 32

PRINT

i/p :- 37

o/p : 98.6

Flow chart :-



6. A math learning application needs to be designed both the quotient and remainder when two numbers are divided. write a prog. to find q, r.

Pseudo code:-

START

Read a, b

quotient = a/b

remainder = $a \% b$

Print

STOP

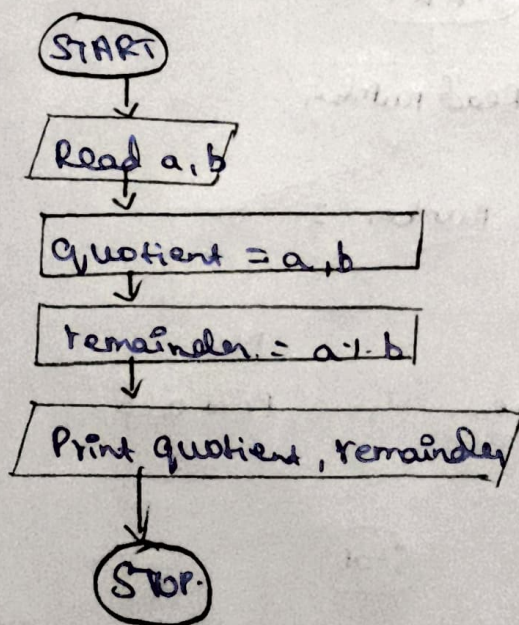
Input:-

$a = 12$, $b = 5$

Output

Quotient = 2 , Remainder = 2

Flow chart



7). An even management system allows entry based on even number tick is . write a c program

Pseudo code :

START

Read number

If Number $\div 2 == 0$ Then

Print "even"

Else

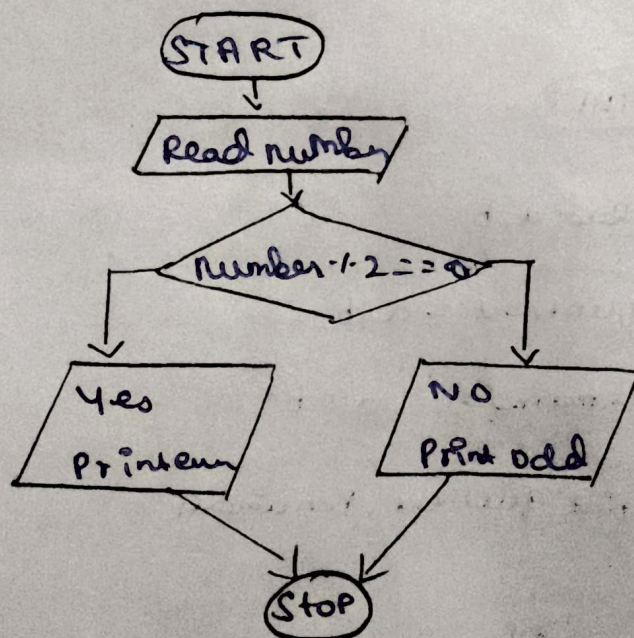
Print "odd"

STOP

Input : 45

Output: Even

Flow chart



8. An engineer need to solve quadratic eqn to form write a c prog to take and display the roots of quadratic equation.

START

Read a, b, c

$$D = b^2 - 4ac$$

IF $D > 0$ THEN

$$\text{root } 1 = (-b + \sqrt{D}) / (2a)$$

$$\text{root } 2 = (-b - \sqrt{D}) / (2a)$$

Print "Real and distinct root"

ELSE IF $D = 0$ THEN

$$\text{root } 1 = -b / (2a)$$

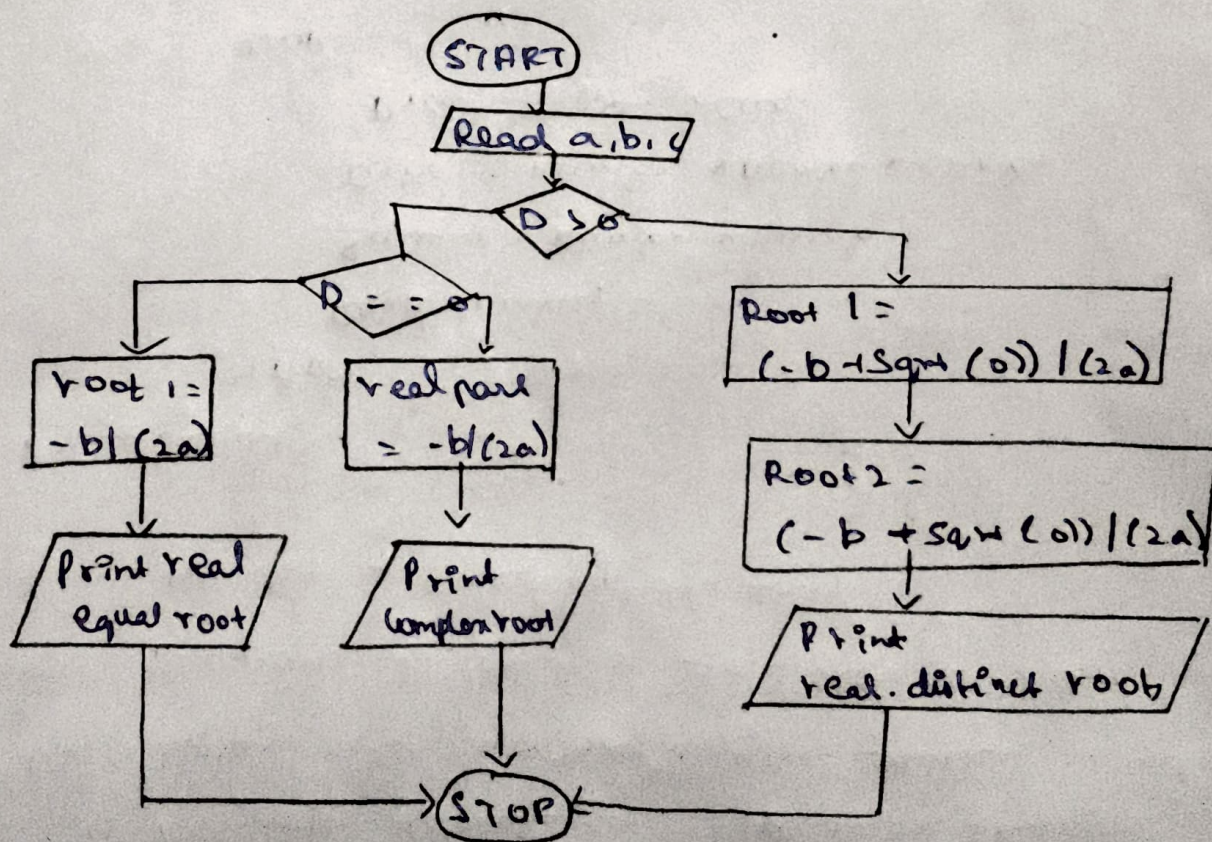
Print "Real and equal roots".

end

STOP

1 | P: $a=1, b=-5, c=6$

0 | P: Real and distinct root.



a). A Scientific Calculated Application requires the square and cube of a given number. write a C program to calculate the square & cube of a number entered by the user

Pseudocode

START

Read number

Square = number * number

Cube = number * number * number

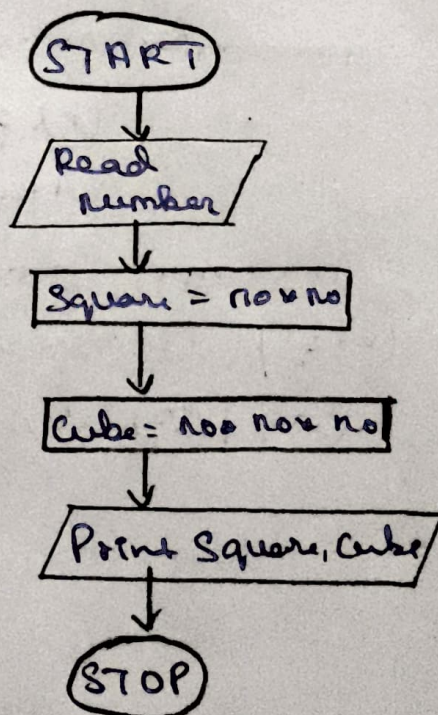
Print Square, Cube

STOP

Input :- 4

Output :- Square = 16 , Cube = 64

Flowchart



10) A company wants to identify the top performer Sales person among three employees on their sales

Pseudocode: START

Read a, b, c

if $a \geq b$ and $a \geq c$ Then

largest = a

else if $b \geq a$ and $b \geq c$

largest = b

Print largest

STOP

Input - $a = 20, b = 45, c = 33$

Output - Largest number = 45

